RIET

Chatbot to chat with medical books using Retrieval-Augmented Generation Model

MedBuddy: Intelligent Support for Medical Education

The Challenge

The Solution

Key Validation

84% Groundedness Score

70% Faster Retrieval

65% Better Efficiency

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Project Team & Guide

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Abstract

Core Technology: Medical students currently face an escalating academic workload and a concerning rise in mental health issues, often struggling to retrieve accurate information from vast, specialized textbooks. Traditional search systems lack contextual medical understanding and frequently fail to provide evidence-based answers. **Technical Implementation:** This project presents MedBuddy, an intelligent digital assistant. To build this, we used the Python 3.12 programming language managed by Miniconda. We used a strategy called **Retrieval-Augmented Generation (RAG)**. In simple terms, RAG is a method where the AI searches for the correct answer in a trusted textbook before responding, rather than just guessing. We utilized **IBM Watsonx APIs,** which provides enterprise-grade, secure Artificial Intelligence tools, replacing less secure public tools.
User Experience: The student interacts with a simple website built with Streamlit. Behind the scenes, the system uses vector embeddings (converting text to numbers) and semantic search to ensure responses are grounded in actual medical content from datasets like PubMed and Meditrin. This ensures the project meets strict security standards required for medical data.

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Introduction

What is LLM (Large Language Model)?

An LLM is a type of Artificial Intelligence that can understand, process, and generate human language. It works like a smart assistant that can read, write, summarize, and answer answer questions.

The Problem with Standard LLMs

Standard models (like basic ChatGPT) work on probability. If they do not know the answer, they might "Hallucinate," which means they confidently make up false medical information.

What is RAG (Retrieval-Augmented Generation)?

RAG is the "Open Book" method. Instead of memorizing everything, the AI looks up the specific page in the textbook **before** answering. This eliminates hallucinations and ensures the answer is true to the source.

Objectives

- To build a grounded AI chatbot.
- To retrieve accurate clinical information from PDF textbooks.
- To generate context-aware answers using IBM Watsonx.
- To improve learning efficiency for medical students.
- To provide grounded medical answers with citations.

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Introduction: Data & Technology

Core Datasets

PubMed: A massive digital library of biomedical literature used to teach the AI general medical language.

Meditrin: A specialized dataset containing specific medical definitions and clinical instructions.

Why IBM Watsonx?

We switched from standard tools to IBM Watsonx because it is 'Enterprise-Grade.' This means it is designed for professional use, ensuring data privacy and using specific models (like Granite-13b-chat) that are trained to follow strict medical instructions.

Key Tools

Streamlit: The tool used to create the website the student sees.

Trulens: A tool used to test how accurate and 'grounded' the AI's answers are.

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Literature Survey (Part 1)

Sl no	Title [Year]	Author	Proposed Method	Disadvantages
1	Chatbot to chat with med books using RAG [2024]	Peri et al.	Base Paper: Integration of RAG and LLMs for PDF interaction.	Initial versions faced high latency (slowness) with multiple large PDFs.
2	Exploring AI Chatbots in Public Health [2023]	Baglivo et al.	Feasibility study on vaccination training bots.	Requires heavy professional verification to avoid misinformation risks.

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Literature Survey (Part 2)

Sl no	Title [Year]	Author	Proposed Method	Disadvantages
3	The Role of ChatGPT in Medical Education [2023]	Gunther Eysenbach	Investigated LLM capabilities in clinical settings.	Raised significant concerns regarding content authenticity and "hallucination."
4	Ontology Based Personal Assistive App [2023]	Geethalekshmy et al.	Semantic reasoning for elderly care.	Focused on personal daily activity logs rather than academic medical education.

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Literature Survey (Part 3)

Sl no	Title [Year]	Author	Proposed Method	Disadvantages
5	Farm-Haystack QA System [2022]	R. S. Nambiar et al.	NLP/backend for pandemic inquiries.	Lacks broad clinical utility; requires domain-specific fine-tuning.
6	ChEMBL Bot [2018]	V. Murali et al.	Conversational interface for chemical data.	Limited only to chemical data queries; cannot provide guidance from clinical textbooks.

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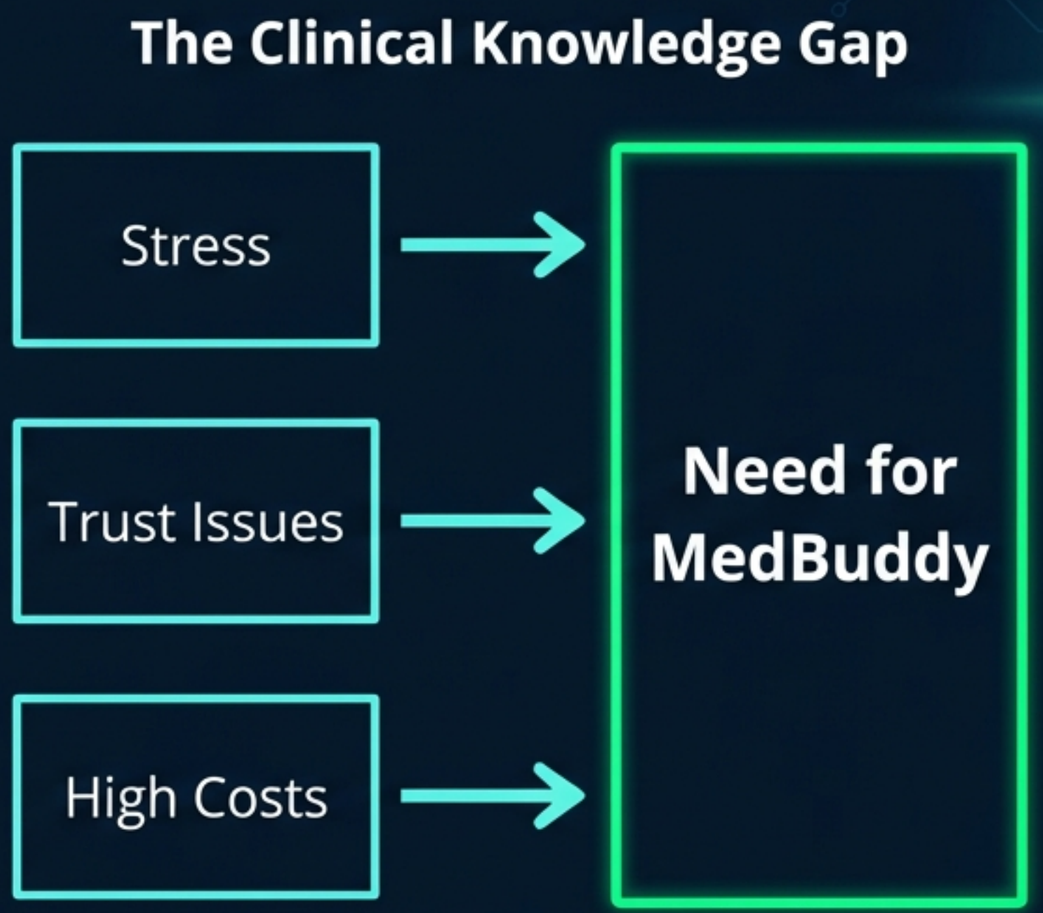
Problem Identification

- **Information Overload:** Medical students face massive workloads. Searching through 1,000+ page PDFs manually is slow and stressful.
- **The Trust Gap (Hallucinations):** Generic AI tools often guess when they don't know the answer. In medicine, a 'guess' is a safety risk.
- **Inefficient Search:** Standard 'Ctrl+F' keyword searches fail to understand the context of complex clinical questions.
- **Resource Inequality:** High-quality, updated medical textbooks are expensive. This limits access for students in resource-constrained areas.



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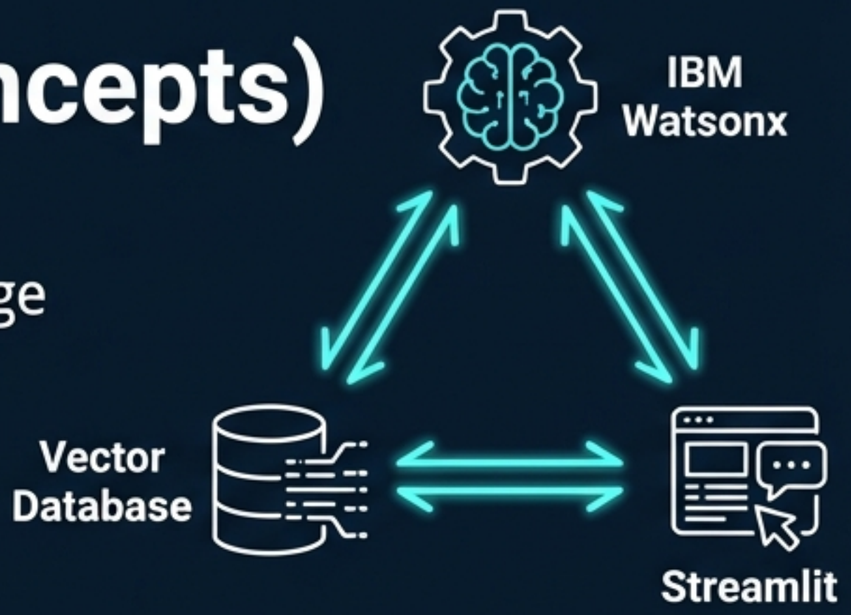
Proposed System (Concepts)



System Overview: MedBuddy integrates IBM Watsonx's Large Language Models with vector embeddings.

Key Concepts:

- **Embeddings:** These are the core of our search. We convert text into long lists of numbers (vectors). Text with similar meanings will have similar numbers, allowing the computer to understand concepts, not just words.
- **Vector Database (ChromaDB):** A special storage system that holds these embeddings. It allows us to find the most relevant paragraph in a massive textbook in milliseconds.
- **Semantic Search:** Unlike a keyword search, this finds matches based on *meaning*. For example, searching 'heart attack' will also find 'myocardial infarction'.

Problem-Solution: Hallucination -> **Grounding (RAG)**. Slow Search -> **Vector Indexing**.

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Proposed System (Implementation)

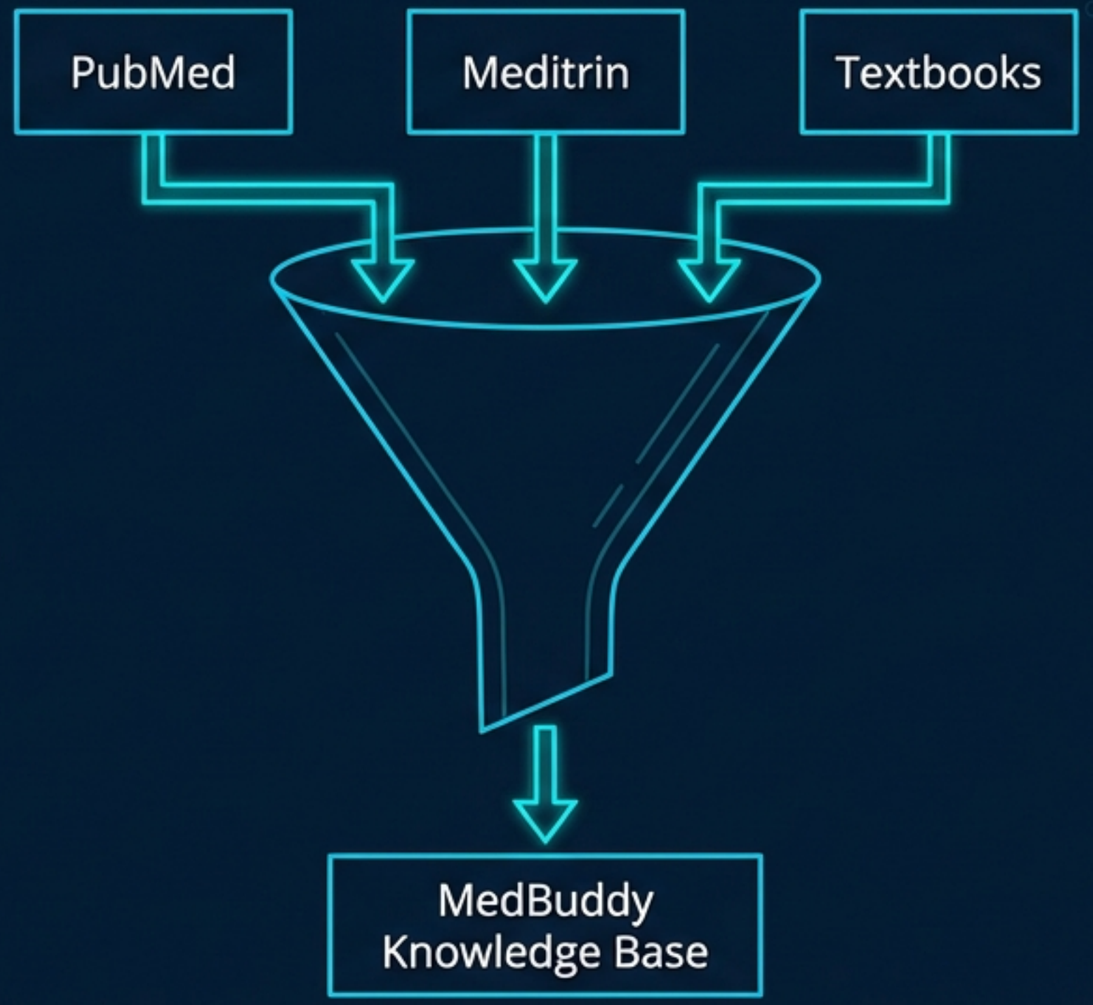
Libraries & Tools:

LlamaIndex: A data framework that acts as the bridge between our PDF data and the AI model. It handles the 'Indexing' step.

HuggingFace Embeddings: A library that provides the 'MiniLM' model to turn text into numbers efficiently on a laptop CPU.

Granite-13b-chat: The specific AI model we use from IBM. It is 'Instruction Tuned," meaning it is trained to follow our command: "Answer strictly using the provided context."

Security: By using local secrets management, we ensure that while we use cloud intelligence, our access keys remain private and secure.



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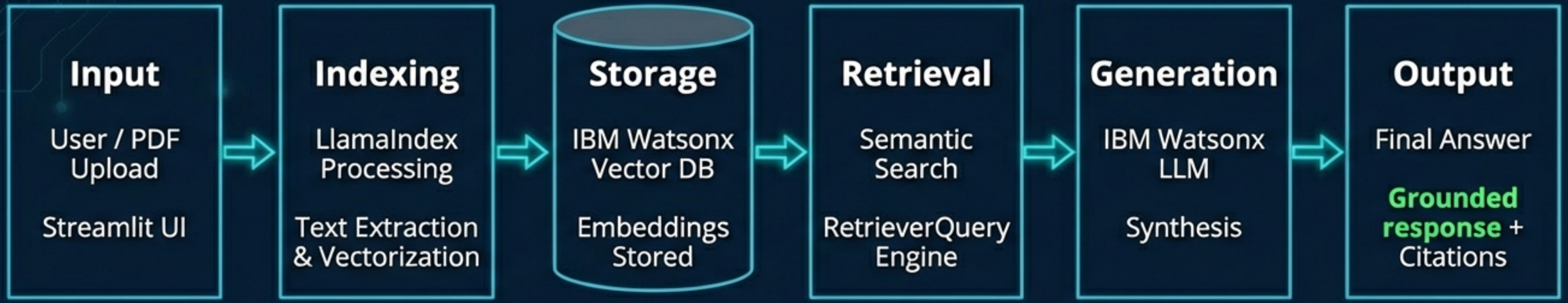
Architecture



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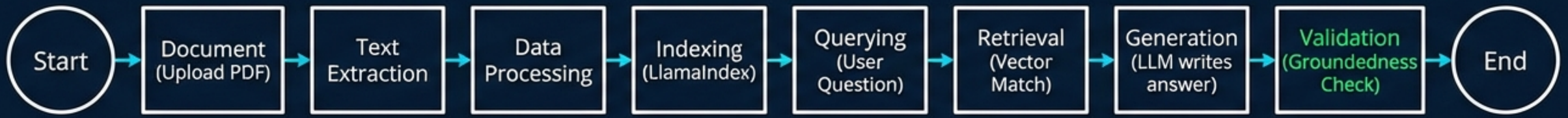
Algorithm



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System and Software Requirements (Hardware)

Laptop Hardware (Minimum):

- **Processor:** Intel Core i5 or equivalent (Optimized for CPU usage).
- **RAM:** 8 GB (To handle the Vector Database).
- **Storage:** 256 GB SSD (For storing PDF datasets).
- **Internet:** Stable connection for IBM Watsonx API calls.

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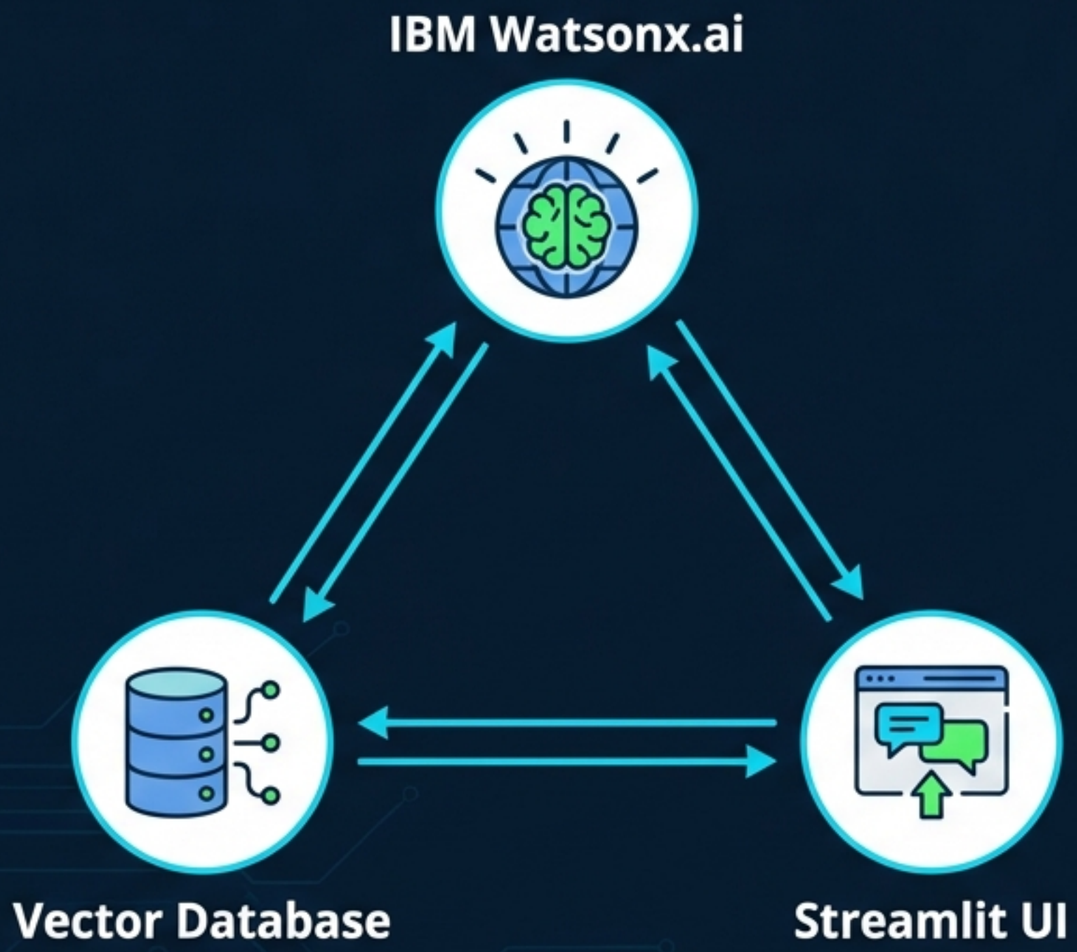
System and Software Requirements (Software)



Front End: Streamlit (Electric Green). A Python library used to build the user interface. It creates the chat box and upload buttons.

Back End: Python 3.12 (Cyan). The core programming language. Managed by **Miniconda** (Electric Green) to keep libraries organized.

AI Engine: IBM Watsonx.ai (Cyan). The cloud platform providing the LLM.

Vector Store: ChromaDB (Green). Stores the mathematical map of the book.

Libraries: PyTorch (Green) (for calculations), **LlamaIndex** (Cyan) (for data connecting).

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Project Plan

Jan 16 2026 - Jan 31 2026
Implement pipeline for PDF upload and vector indexing.

Data Ingestion.
Data Ingestion.
↳ PDF Ingestion & Preprocessing

Feb 01 2026 - Feb 15 2026
Connect IBM Watsonx APIs and establish the retrieval model.

RAG Integration.
RAG Integration.
↳ RetrieverQueryEngine.query

Feb 16 2026 - Feb 28 2026
Build the conversational interface using Streamlit.

UI Development.
UI Development.
↳ CompactAndRefine.get_response

Mar 01 2026 - Mar 10 2026
Test for groundedness (target 84%) and security (HIPAA compliance).

Evaluation.
Evaluatt on secure cloud infrastructure.
↳ LLMPredictor.predict

Mar 15 2026
Launch on secure cloud infrastructure.

Final Deployment.

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Future Plan

Research Acceleration

Rapidly summarize findings from thousands of new research papers.

Clinical Decision Support

AI co-pilots for doctors, providing real-time access to treatment guidelines.

Multimodal Capabilities

Ability to interpret medical images (like X-rays) and accept voice commands.

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Conclusion

84%
Groundedness Score (High reliability)

70%
Reduction in retrieval time

65%
Improvement in learning efficiency

Summary: MedBuddy successfully addresses the problem of information overload in medical education.

Impact: By democratizing access to medical knowledge, MedBuddy fosters a more inclusive and equitable educational landscape, reducing stress and improving learning outcomes for students.

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References

1. Peri et al., 'Chatbot to chat with medical books using Retrieval-Augmented Generation Model,' IEEE, 2024.
2. F. Baglivo et al., 'Exploring the Possible Use of AI Chatbots in Public Health Education,' JMIR Publications, 2023.
3. G. Eysenbach, 'The Role of ChatGPT and AI in Medical Education,' JMIR Publications, 2023.
4. N. V. Rohil and A. Nair, 'Ontology Based Personal Assistive Application,' IEEE Global Conference for Advancement in Technology, 2023.
5. R. S. Nambiar et al., 'Farm-Haystack Question Answering System,' International Conference on Cloud Computing, 2022.
6. V. Murali et al., 'ChEMBL Bot - A Chat Bot for ChEMBL Database,' International Conference on Information Processing, 2018.

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